Bridging the Simulation-to-Reality Gap:

A Comparative Study of Robotics Simulation Platforms

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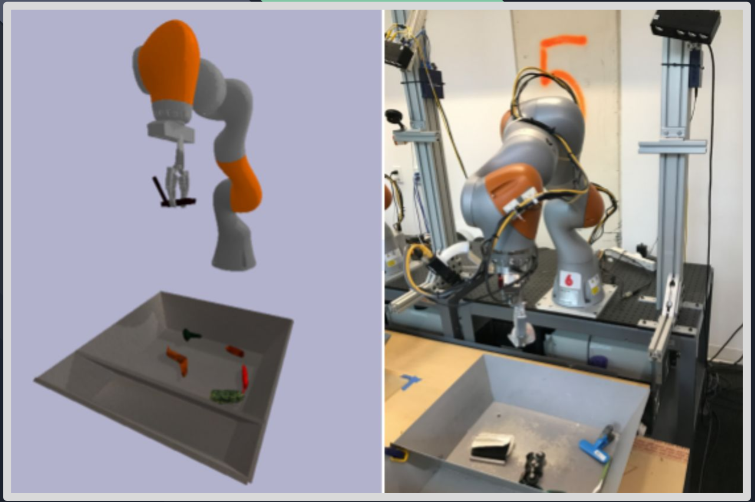
Introduction



Simulated robot manipulation task (left) vs real robot execution (right) - (Google Research, 2020) [2]

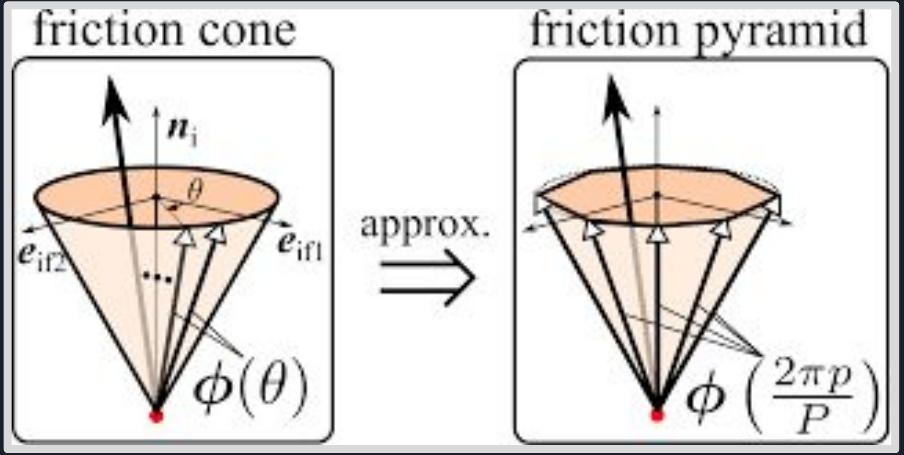
Simulation is the imitation of a real-world process or system created in a virtual environment [1].

- Widely used in robotics for safe, low-cost testing.
- Different simulators can produce different results, even with the same robot model.

Background: Simulation-to-Reality Gap

Physics-based simulators approximate real-world robot dynamics.

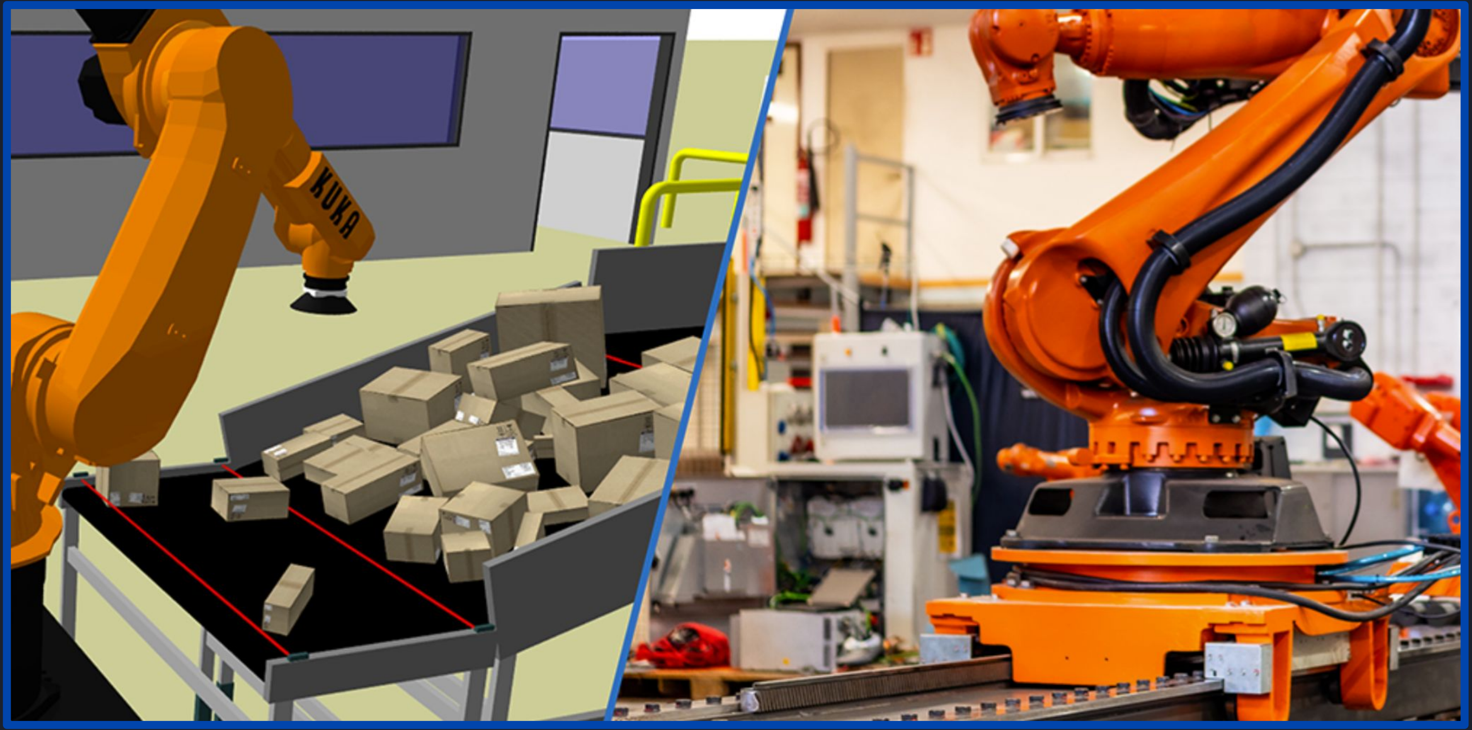
Each physics engine models contact, friction, and joint motion differently.



Friction cone (left) and simplified friction pyramid approximation (right), used in many physics engines for contact and friction modeling [3].

[4][5][6][7]

Applications + Importance of Bridging the Gap



Digital Twin: Simulated robot environment (left) vs. real robot system (right) - KUKA Robotics (2023) [8]

Objective

Compare how different physics-based simulation platforms model robot motion and physical interactions using identical robot models and standardized tests.

Simulators:

Methodology

- Identical URDF models (UF850, xArm6)
- Two standardized test scenarios
- Consistent physics settings
- Focus platforms: Gazebo, Unity

UFactory xArm 6 [9]

UFactory UF850 [10]

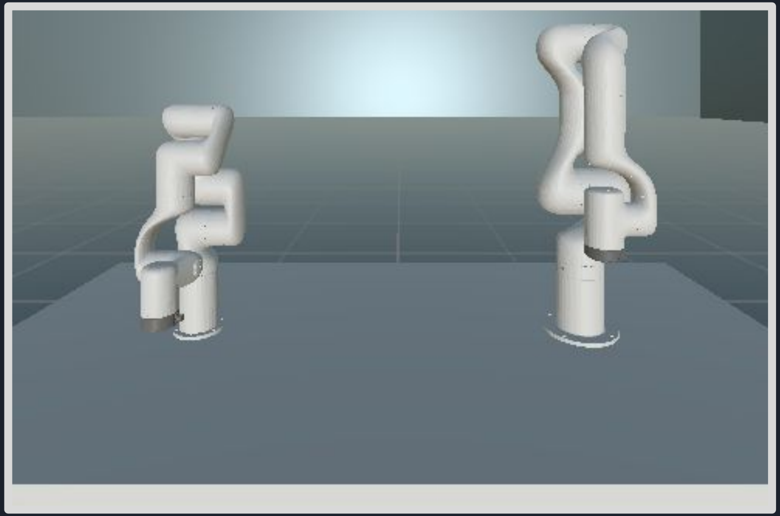
Test Scenarios

Test 1 — Joint Trajectory Tracking
Evaluates how accurately simulators follow commanded robot motion.

Test 2 — Friction & Sliding Test
Measures how simulators model basic physics like friction and acceleration.

Test 1: Trajectory Tracking

- Predefined joint trajectories executed on UF850 and xArm6
- Identical commands sent to each simulator
- Logged: time, commanded angles, actual angles, joint speeds

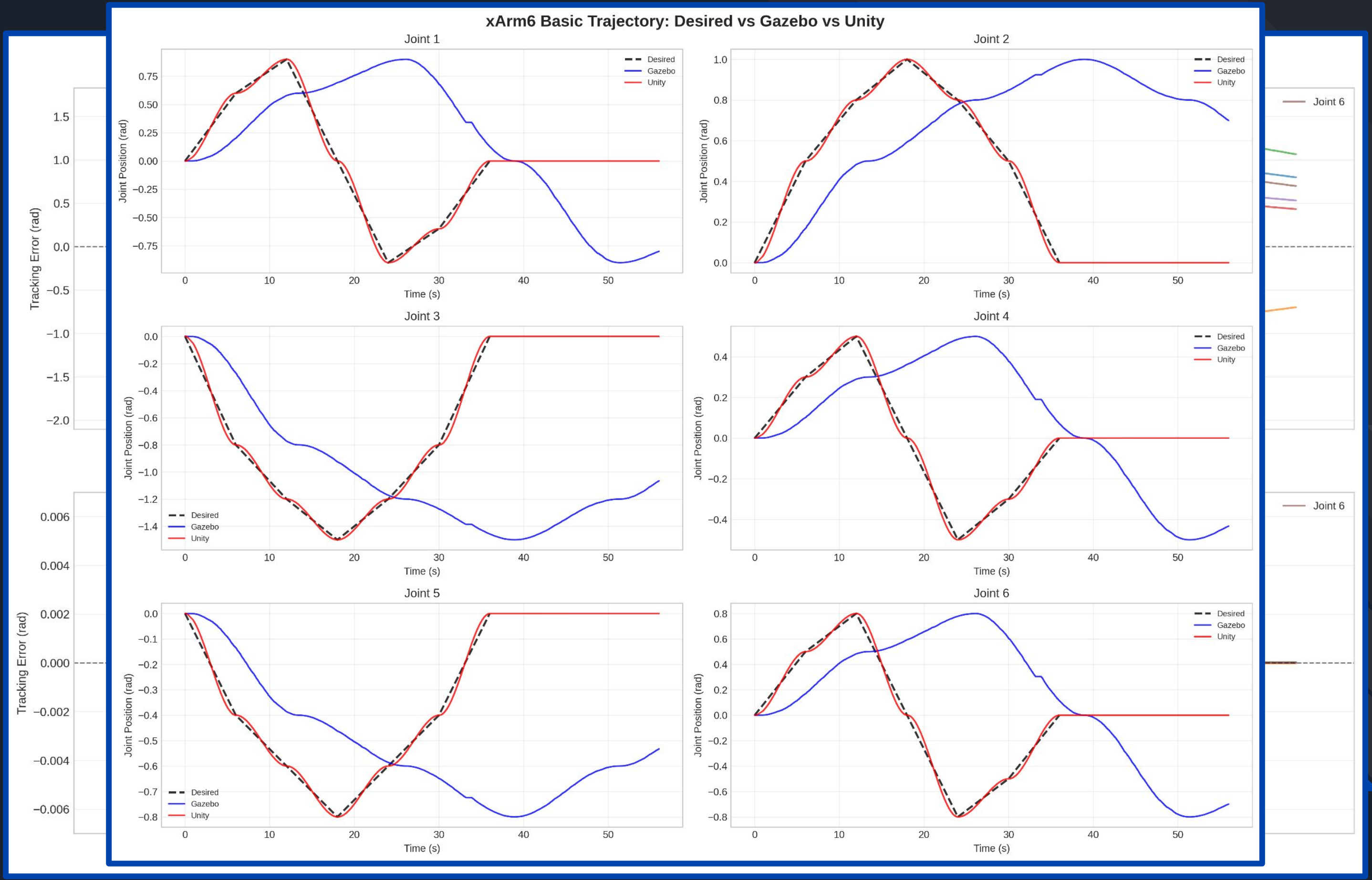


Unity Environment for Trajectory Tracking Test

Test 1 Results

Complex Trajectory Tracking Test on Gazebo with UF850 Robot Model (2x speed)

Test 1 Results

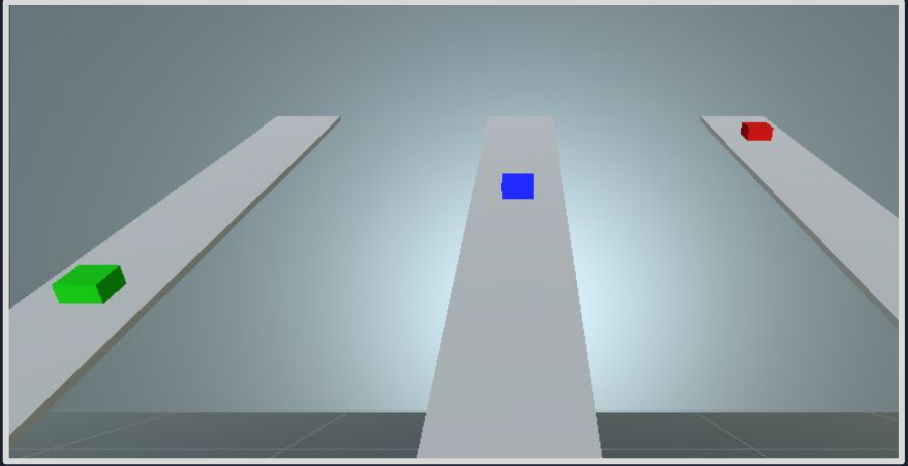


Test 1: Results Summary

Test (Robot & Trajectory)	Gazebo Mean Error (°)	Unity Mean Error (°)	Gazebo Max Error (°)	Unity Max Error (°)
UF850 Basic	10.574	0.023	23.012	13.411**
UF850 Complex	29.55	0.065	77.16	19.408**
xArm6 Basic	31.534	0.08	72.968	0.194
xArm6 Complex	36.755	0.149	103.313	0.534

Test 2: Friction and Sliding

- 1 kg box on a 30° incline
- Three friction levels: $\mu = 0.1, 0.3, 0.5$
- Identical geometry, mass, and physics parameters
- Logged: time, position, speed, stopping distance, duration

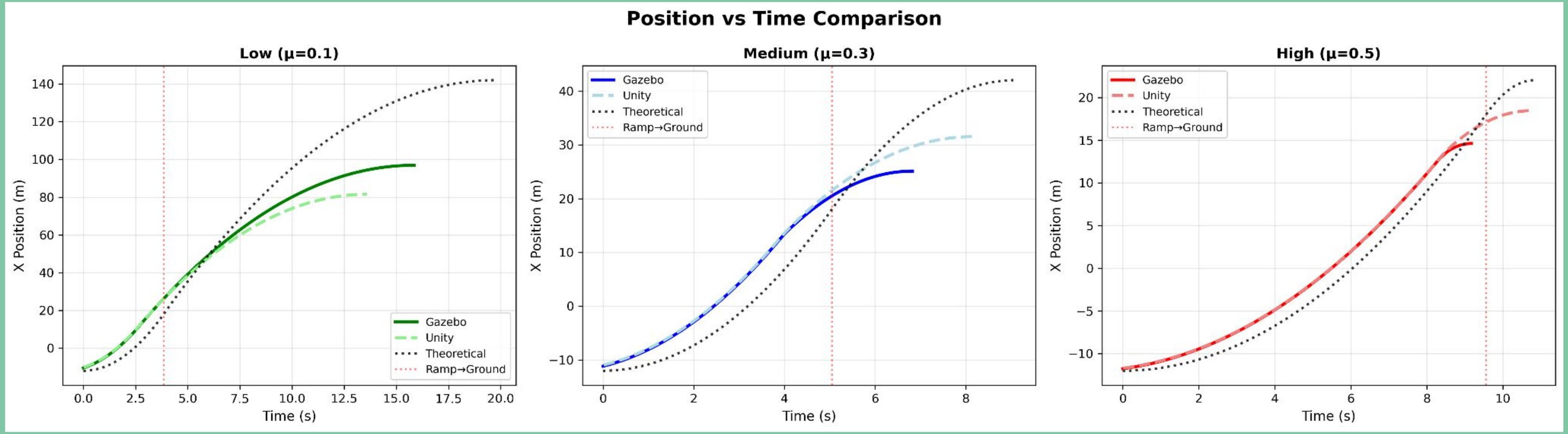


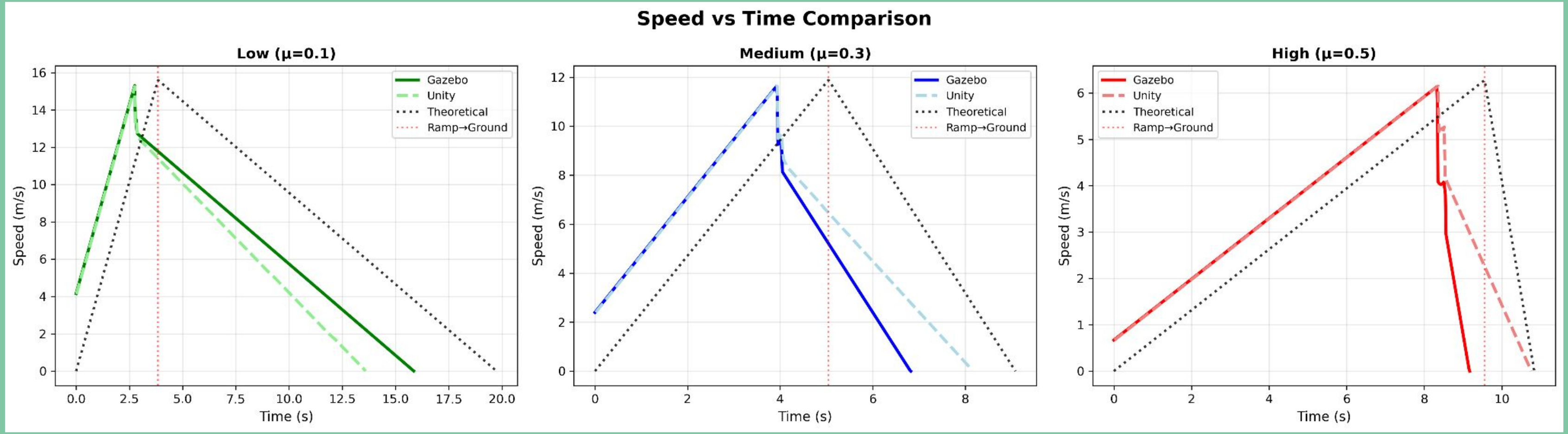
Unity Friction Test Setup for Three Friction Conditions

Test 2 Results

Unity Simulation of Low (green), Medium (blue), and High (red) Friction Conditions

Test 2 Results





Test 2: Results Summary

Friction Level	Platform	Speed Error (%)	Distance Error (%)	Duration Error (%)
Low (μ = 0.1)	Gazebo	1.89	30.24	19.73
	Unity	2.25	40.47	31.15
Medium (μ = 0.3)	Gazebo	2.22	32.87	24.89
	Unity	2.1	21.4	10.24
High (μ = 0.5)	Gazebo	2.26	22.5	15.38
	Unity	2.05	11.4	0.73

Conclusions

- Simulators produced noticeably different motion and physics behavior under identical tests.
- Unity tracked joint trajectories more accurately; Gazebo showed larger deviations.
- Friction tests showed both engines follow general trends but differ quantitatively from theory.
- Characterizing these differences helps interpret simulation results and improves transferability.

Future Work

- Perform Isaac Sim implementation and benchmarking
- Compare simulated results with real robot data
- Improve parameter tuning
- Extend tests to more complex interactions and control tasks

Thank you for watching

References

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